

Consumption-Saving

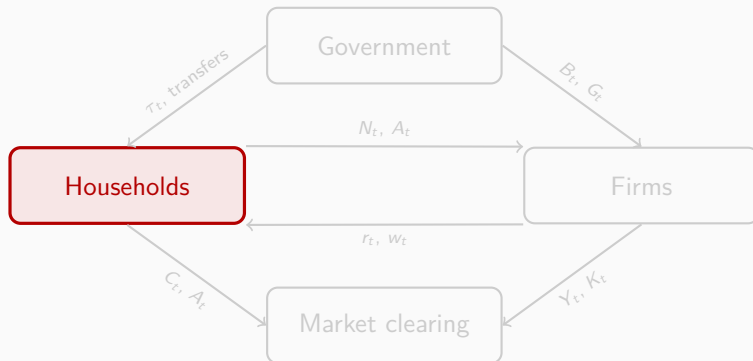
Adv. Macro: Heterogeneous Agent Models

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Introduction

We open the household block



This course focuses on the household block.

The question: *when does the distribution matter for aggregates?*

Goals for today

- **Review consumption-saving theory**
 1. The permanent income hypothesis, and what it gets wrong
 2. The buffer-stock model: risk and borrowing constraints
- **Solve the household problem in partial equilibrium**
 1. Basics of dynamic programming
 2. Value function iteration (VFI)
 3. The endogenous grid method (EGM)
- **Simulate the distribution and aggregate**
 1. Monte Carlo
 2. The histogram method (Young, 2010)
- **Note:** partial equilibrium for now, prices are *fixed*. They become equilibrium objects in the next lecture.

- **Generations of models:**
 1. **Permanent income hypothesis (PIH)** (Friedman, 1957)
or life-cycle model (Modigliani and Brumberg, 1954)
 2. **Buffer-stock consumption model**
Deaton (1991, 1992); Carroll (1992, 1997, 2019)
 3. **Multiple-asset buffer-stock consumption models**
e.g. Kaplan and Violante (2014); Harmenberg and Öberg (2021)
 - **Consumption-and-saving over the life-cycle dynamic**
e.g. Gourinchas and Parker (2002); Druedahl and Martinello (2022)
 - **Empirical MPCs and income risk**
e.g. Fagereng et al. (2021); Guvenen et al. (2021)
- Book:** The Economics of Consumption, Jappelli and Pistaferri (2017)

1. Introduction
2. PIH
3. Dynamic programming
4. Buffer-stock
5. VFI applied to the Buffer Stock Model
6. EGM
7. Simulation and aggregation
8. Recap
9. Misc

PIH



- **Keynesian consumption function:** $C_t = \bar{C} + bY_t$ Consumption responds to *current* income. Multiplier is $\frac{1}{1-b}$.
But it lacks micro-foundations!
- **Permanent income hypothesis** (Friedman, 1957): Consumption responds to the *present value* of income.
Current income matters only through it.
- But: PIH implies too low MPCs. We will see how to fix it

$$v_0 = \max_{\{c_t\}_{t=0}^{T-1}} \sum_{t=0}^{T-1} \beta^t u(c_t)$$

s.t.

$$a_t = (1 + r)a_{t-1} + wz_t - c_t$$

$$a_{T-1} \geq 0, \quad a_{-1} \text{ given}$$

- **Variables:** Consumption: c_t Productivity: z_t End-of-period savings: a_t (*no debt at death*)
- **Parameters:** Discount factor: β Wage: w Interest rate: r (define $R \equiv 1 + r$ as interest factor)

- **Only the terminal condition really constrains:** without $a_{T-1} \geq 0$ the household would just borrow more and more
- **Substitution** implies *Intertemporal Budget Constraint* (IBC)

$$\begin{aligned}
 a_{T-1} &= Ra_{T-2} + wz_{T-1} - c_{T-1} \\
 &= R(Ra_{T-3} + wz_{T-2} - c_{T-2}) + wz_{T-1} - c_{T-1} \\
 &= R^T a_{-1} + \sum_{t=0}^{T-1} R^{T-1-t} (wz_t - c_t)
 \end{aligned}$$

- Use **terminal condition** $a_{T-1} = 0$ (equality due to utility max.)

$$R^{-(T-1)} a_{T-1} = 0 \Leftrightarrow \sum_{t=0}^{T-1} R^{-t} c_t = s_0 + h_0$$

where $s_0 \equiv Ra_{-1}$ (after-interest assets) and $h_0 \equiv \sum_{t=0}^{T-1} R^{-t} wz_t$ (human capital)

$$\mathcal{L} = \sum_{t=0}^{T-1} \beta^t u(c_t) + \lambda \left[s_0 + h_0 - \sum_{t=0}^{T-1} R^{-t} c_t \right]$$

- It is a **static** problem: 1. **Information:** s_0 and h_0 are known at $t = 0$ 2. **Target:** Discounted utility, $\sum_{t=0}^{T-1} \beta^t u(c_t)$ 3. **Behavior:** Choose $c_0, c_1, \dots, c_{T-1}$ *simultaneously* 4. **Solution:** Sequence of consumption *choices* $c_0^*, c_1^*, \dots, c_{T-1}^*$ - **First order conditions:**

$$\forall t : 0 = \beta^t u'(c_t) - \lambda R^{-t} \Leftrightarrow u'(c_t) = \lambda (\beta R)^{-t}$$

- **Euler-equation** for $k \in \{1, 2, \dots\}$:

$$\frac{u'(c_t)}{u'(c_{t+k})} = \frac{\lambda (\beta R)^{-t}}{\lambda (\beta R)^{-(t+k)}} = (\beta R)^k$$

Consumption choice

- **CRRA:** $u(c_t) = \frac{c_t^{1-\sigma}}{1-\sigma}$ implies the Euler-equation

$$\frac{c_0^{-\sigma}}{c_t^{-\sigma}} = (\beta R)^t \Leftrightarrow c_t = (\beta R)^{\frac{t}{\sigma}} c_0$$

Constant consumption if:

1. $\beta R = 1$
 2. $\frac{1}{\sigma} \rightarrow 0$ (zero intertemporal elasticity of substitution)
- Insert **Euler** into **IBC** to get consumption choice

$$\sum_{t=0}^{T-1} \left((\beta R)^{1/\sigma} R^{-1} \right)^t c_0 = (s_0 + h_0) \Leftrightarrow$$

$$c_0^* = \frac{1 - (\beta R)^{1/\sigma} R^{-1}}{1 - ((\beta R)^{1/\sigma} R^{-1})^T} (s_0 + h_0)$$

Infinite horizon: $T \rightarrow \infty$

- **Infinite horizon** for $(\beta R)^{1/\sigma} R^{-1} < 1$: Let $T \rightarrow \infty$ to get

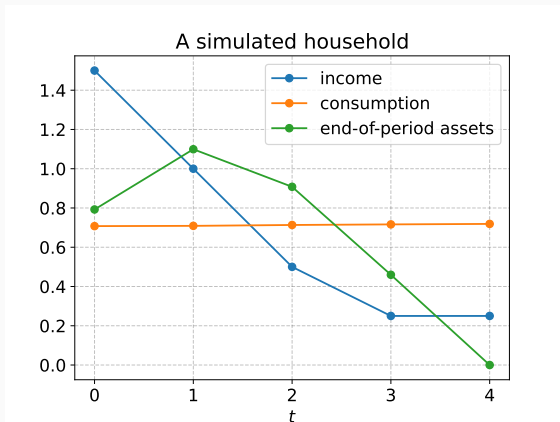
$$c_0^* = \left(1 - \frac{(\beta R)^{1/\sigma}}{R}\right) (s_0 + h_0)$$

Assuming $\beta R = 1$, this becomes

$$c_0 = \frac{r}{1+r} (s_0 + h_0)$$

$\Rightarrow$ Consumption is the annuity value of total wealth (perfect consumption smoothing!)

Consumption smoothing



Income falls by two thirds, consumption stays almost constant.

The household saves while income is high and runs the savings down when it is low. ($T = 3$, $a_{-1} = 0$, from 02_vfi_pih.ipynb)

Propensities to consume ($\beta R = 1$, $z_t = 1$)

Assume $\beta R = 1$ and $z_t = 1$, then

$$c_0^* = ra_{-1} + w$$

MPC for different types of shocks (exact derivatives):

1. MPC of *windfall* income: $\frac{\partial c_0}{\partial s_0} = \frac{r}{1+r} \approx r$
2. MPC of *future* income change: $\frac{\partial c_0}{\partial w z_t} = \frac{r}{1+r} (1+r)^{-t} \rightarrow$ even smaller
3. MPC of *permanent* income change: $\frac{\partial c_0}{\partial w} = 1$

Connection to the representative agent

- The consumption function is *linear* in resources

$$c_0^* = \kappa (s_0 + h_0), \quad \kappa \equiv 1 - \frac{(\beta R)^{1/\sigma}}{R}$$

with the *same* κ for every household

- **Exact aggregation:** sum over households i

$$\sum_i c_{0,i}^* = \kappa \left(\sum_i s_{0,i} + \sum_i h_{0,i} \right)$$

- **Implication:** aggregate consumption depends on *aggregate* wealth and *aggregate* income only. The distribution does not enter, and a representative agent gets the aggregates exactly right.
- **This is why the course exists:** *heterogeneity matters for aggregates only once this linearity breaks.*

Dynamic programming

Derive the Bellman equation

Rewrite the consumption savings problem

$$\begin{aligned}v_t(a_{t-1}) &= \max_{\{c_s\}_{s=t}^{T-1}} \sum_{s=t}^{T-1} \beta^{s-t} u(c_s) \\&= \max_{\{c_s\}_{s=t}^{T-1}} \left[u(c_t) + \beta \sum_{s=t+1}^{T-1} \beta^{s-t-1} u(c_s) \right] \\&= \max_{c_t} \left[u(c_t) + \beta \underbrace{\max_{\{c_s\}_{s=t+1}^{T-1}} \left[\sum_{s=t+1}^{T-1} \beta^{s-(t+1)} u(c_s) \right]}_{=v_{t+1}(a_t)} \right] \\&= \max_{c_t} \left[u(c_t) + \beta v_{t+1}(a_t) \right]\end{aligned}$$

This is called the **Bellman equation**.

Dynamic Programming vs Sequence Problem

$$v_t(a_{t-1}) = \max_{c_t} \left[u(c_t) + \beta v_{t+1}(a_t) \right]$$

Main differences:

1. Now we maximize only with respect to c_t , not the entire path!
2. We obtain $c_t^*(a_{t-1})$, which is called a **policy function**
3. $v_{t+1}(a_t)$ is called a **value function**.

Solving the problem amounts to finding those two functions! How can we do it?

Backward induction

To solve the problem in finite horizon, we use **backward induction**.

Let's start from the end. In the last period, $t = T - 1$, we have

$$v_{T-1}(a_{T-2}) = \max_{c_{T-1}} u(c_{T-1}) + \beta v_T(a_{T-1})$$

There is no period T , so $v_T(a_{T-1}) = 0$, and with $a_{T-1} \geq 0$ the household consumes everything, $c_{T-1} = (1 + r)a_{T-2} + wz_{T-1}$. Thus:

$$v_{T-1}(a_{T-2}) = u((1 + r)a_{T-2} + wz_{T-1})$$

We can then go backward in time, and solve $v_{T-2}(a_{T-3})$, etc, until $v_0(a_{-1})$.

Euler equation and envelope condition

For $t < T - 1$, substitute the budget constraint into the objective

$$v_t(a_{t-1}) = \max_{a_t} \left[u((1+r)a_{t-1} + wz_t - a_t) + \beta v_{t+1}(a_t) \right]$$

The first-order condition is $u'(c_t) = \beta v'_{t+1}(a_t)$.

To obtain v'_{t+1} , differentiate with respect to the *state*, a_{t-1} :

$$\begin{aligned} v'_t(a_{t-1}) &= u'(c_t) \left(1 + r - \frac{\partial a_t}{\partial a_{t-1}} \right) + \beta v'_{t+1}(a_t) \frac{\partial a_t}{\partial a_{t-1}} \\ &= u'(c_t)(1+r) + \underbrace{\frac{\partial a_t}{\partial a_{t-1}} \left(u'(c_t) - \beta v'_{t+1}(a_t) \right)}_{=0} = u'(c_t)(1+r) \end{aligned}$$

This is the **envelope theorem**!

Shift one period forward and insert in the FOC: $u'(c_t) = \beta R u'(c_{t+1})$.

Same Euler-equation as in the sequence problem, hence same solution!

A dynamic programming “recipe”

Start at $t = T - 1$ with $v_T(\bullet) = 0$. For every possible a_{t-1} :

1. Maximize $u(c_t) + \beta v_{t+1}(a_t)$ to get $c_t^*(a_{t-1})$
2. Compute $v_t(a_{t-1}) = u(c_t^*(a_{t-1})) + \beta v_{t+1}(a_t^*)$

Go backward in time, until $t = 0$.

We have obtained a sequence of policy functions $c_t^*(a_{t-1})$ and value functions $v_t(a_{t-1})$!

We need three computational tools

To do this on a computer, we need a couple of things:

1. How do we loop over all possible a_{t-1} ? → **Create a grid**
2. We don't have an analytical expression for $v_{t+1}(a_t)$. How do we approximate it? → **Use linear interpolation**
3. How do we maximize a function? → **grid search or golden section search!**

Let's code!

Infinite Horizon: Convergence

What about the infinite horizon, $T \rightarrow \infty$?

Under some regularity conditions, it turns out that $v_t(a_{t-1}) \rightarrow v(a_{t-1})$

$\Rightarrow$ The problem becomes **stationary**, that is, it does not depend on time.

Infinite Horizon: Value Function Iteration

How can we use this computationally?

Start from a guess $v^0(a_{t-1})$, e.g. $v^0 = 0$ as in the last period of life.

Then, given the current guess v^n , for all a_{t-1} :

- Find optimal consumption c_t^* by maximizing $u(c_t) + \beta v^n(a_t)$
- Update the guess on the value function,
$$v^{n+1}(a_{t-1}) = u(c_t^*) + \beta v^n(a_t^*)$$

Iterate until $\max |v^{n+1} - v^n| < \epsilon$. This is **value function iteration** (VFI).

We will sometimes call this the **backward step**: given a continuation value, what is optimal today.

Let's code!

Buffer-stock

MPCs in the data are high

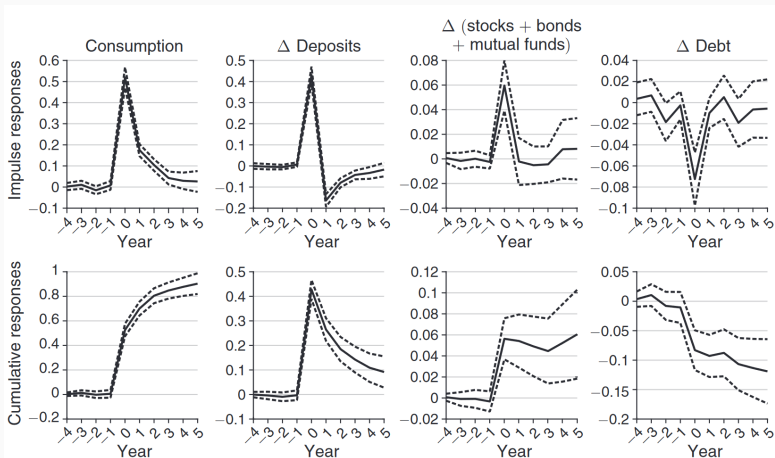


FIGURE 2. DYNAMIC HOUSEHOLD RESPONSES TO LOTTERY PRIZES

Source: Fagereng et al. (2021). **PIH says** $\frac{r}{1+r} \approx 2\%$.

What is missing from the model

We make two changes to the PIH model

1. Introduce uninsurable **income risk**
2. A **borrowing constraint**

This gives us:

1. A precautionary savings motive (on top of consumption smoothing)
2. Higher MPCs when households are close to the borrowing constraint
3. Break the linearity of the consumption function $\Rightarrow$ distribution starts to matter

Uncertainty and a borrowing constraint

$$v_0(z_0, a_{-1}) = \max_{\{c_t\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right]$$

s.t.

$$a_t = (1 + r)a_{t-1} + wz_t - c_t \geq \underline{a}$$

$$\log z_t = \rho_z \log z_{t-1} + \psi_t, \quad \psi_t \sim \mathcal{N}(0, \sigma_\psi^2)$$

$$\lim_{t \rightarrow \infty} (1 + r)^{-t} a_t \geq 0 \quad [\text{No-Ponzi game}]$$

A true dynamic problem:

1. **Information:** z_t is revealed period-by-period
2. **Target:** *Expected* discounted utility, $\mathbb{E}_0 [\sum_{t=0}^{\infty} \beta^t u(c_t)]$
3. **Behavior:** Choose c_t *sequentially* as information is revealed
4. **Solution:** Consumption *function*, $c^*(z_t, a_{t-1})$

Why is the sequential formulation difficult?

- With uncertainty, the optimal sequence is conditional on a given realization of shocks
- An optimal sequence should be optimal for every potential realizations of shocks!
- Even if z_t takes only $\#_z$ values, there are $\#_z^T$ possible histories after T periods, each with its own consumption path $\Rightarrow$ **Unfeasible!**

This is where **dynamic programming** becomes necessary

$$\begin{aligned} v(z_t, a_{t-1}) &= \max_{c_t} u(c_t) + \beta \mathbb{E}_t [v(z_{t+1}, a_t)] \\ \text{s.t. } a_t &= (1 + r)a_{t-1} + wz_t - c_t \geq \underline{a} \end{aligned}$$

1. **State variables:** z_t and a_{t-1}
2. **Control (choice) variable:** c_t
3. **Continuation value:** $\beta \mathbb{E}_t [v(z_{t+1}, a_t)]$
4. **Parameters:** r , w , and stuff in $u(\bullet)$

Note: no time subscript on v . The problem is stationary, so the policy function is the same in every period.

- **Substitution** still implies (path-by-path, for any realization of z_t):

$$R^{-(T-1)}a_{T-1} = 0 \Leftrightarrow \sum_{t=0}^{T-1} R^{-t}c_t = s_0 + h_0$$

- **What if $T \rightarrow \infty$?** We must have $\lim_{T \rightarrow \infty} R^{-(T-1)}a_{T-1} = 0$
 1. $\lim_{T \rightarrow \infty} R^{-(T-1)}a_{T-1} > 0$: Consumption can be increased
 2. $\lim_{T \rightarrow \infty} R^{-(T-1)}a_{T-1} < 0$: Violates No-Ponzi game condition
- For $T \rightarrow \infty$ we have the **IBC**:

$$\sum_{t=0}^{\infty} R^{-t}c_t = Ra_{-1} + \sum_{t=0}^{\infty} R^{-t}wz_t$$

Natural borrowing limit

What is the lowest possible value of $\underline{a}$?

- Denote **minimum possible productivity** by $\underline{z}$
- **Consumption must be non-negative** $\Rightarrow$
interest payments must be less than minimum income

$$c_t \geq 0 \Rightarrow r(-a_t) \leq w\underline{z} \Leftrightarrow a_t \geq -\frac{w\underline{z}}{r}$$

If debt was larger it would in the worst case ($\forall z_t = \underline{z}$) grow without bound even with zero consumption ($\forall c_t = 0$)

$$a_0 = -\frac{w\underline{z}}{r} - \Delta$$

$$a_1 = (1+r)a_0 + w\underline{z} = -\frac{w\underline{z}}{r} - (1+r)\Delta$$

$$a_2 = (1+r)a_1 + w\underline{z} = -\frac{w\underline{z}}{r} - (1+r)^2\Delta$$

$\vdots$

Euler-equation with an occasionally binding constraint

- Write the **Lagrangian**

$$\mathcal{L} = \frac{c_t^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}_t [v(z_{t+1}, a_t)] + \mu_t (a_t - \underline{a})$$

Then:

$$\text{FOC: } c_t^{-\sigma} = \beta \mathbb{E}_t [v_a(z_{t+1}, a_t)] + \mu_t$$

$$\text{Slackness: } \mu_t (a_t - \underline{a}) = 0$$

$$\text{Envelope: } v_a(z_t, a_{t-1}) = (1+r)c_t^{-\sigma}$$

- Two cases:**

- Unconstrained** ($a_t > \underline{a}$, so $\mu_t = 0$):

$$c_t^{-\sigma} = \beta R \mathbb{E}_t [c_{t+1}^{-\sigma}]$$

- Constrained** ($a_t = \underline{a}$, so $\mu_t > 0$):

$$c_t^{-\sigma} > \beta R \mathbb{E}_t [c_{t+1}^{-\sigma}] \quad \text{and} \quad c_t = (1+r)a_{t-1} + wz_t - \underline{a}$$

Prudence and precautionary saving

- **Prudence:** $u'''(c) > 0$, i.e. marginal utility is *convex* CRRA has $u'''(c) = \sigma(\sigma + 1)c^{-\sigma-2} > 0$
- **Jensen on the Euler-equation** (take $\beta R = 1$, unconstrained):

$$u'(c_t) = \mathbb{E}_t[u'(c_{t+1})] > u'(\mathbb{E}_t[c_{t+1}]) \Leftrightarrow \mathbb{E}_t[c_{t+1}] > c_t$$

- **Interpretation:** the household *tilts consumption into the future* to build a buffer. Risk lowers consumption today even with no constraint.
- **Two distinct channels, same direction:**
 1. **Risk $\times$ prudence:** precautionary saving
 2. **Borrowing constraint:** cannot smooth downwards

Special case I: Quadratic utility

- **Quadratic utility:** $u(c_t) = -\frac{1}{2}(\bar{c} - c_t)^2$ with $\beta R = 1$ and “large” $\bar{c}$
- **Euler-equation:** *Consumption = expected future consumption*

$$(\bar{c} - c_t) = \mathbb{E}_t[(\bar{c} - c_{t+k})] \Leftrightarrow c_t = \mathbb{E}_t[c_{t+k}]$$

- Use **IBC** in expectation to get **consumption function**:

$$\sum_{s=0}^{\infty} R^{-s} \mathbb{E}_0[c_s] = Ra_{-1} + \sum_{s=0}^{\infty} R^{-s} w \mathbb{E}_0[z_s] \Rightarrow$$
$$c^*(z_0, a_{-1}) = c_0 = ra_{-1} + \frac{r}{R} \sum_{s=0}^{\infty} R^{-s} w \mathbb{E}_0[z_s]$$

where we formally disregard the borrowing constraint

- **Certainty equivalence:** *Only expected income matters.* No prudence: $u''' = 0$.

Special case II: CARA utility

- **CARA utility:** $u(c_t) = -\frac{1}{\alpha} e^{-\alpha c_t}$
- **Productivity is absolute random walk:**

$$z_t = z_{t-1} + \psi_t, \quad \psi_t \sim \mathcal{N}(0, \sigma_\psi^2)$$

- **Consumption function** (see proof):

$$c^*(z_t, a_{t-1}) = ra_{t-1} + wz_t - \frac{\frac{1}{\alpha} \log(\beta R) + \frac{\alpha w^2 \sigma_\psi^2}{2}}{r}$$

where we formally disregard the borrowing constraint

- **Precautionary saving:** $\sigma_\psi^2 \uparrow$ implies $c_t^* \downarrow$ for given z_t and a_{t-1}
 $\Rightarrow$ *accumulation of buffer-stock*
- **But:** the MPC is still *constant*. To solve the problem with a more interesting utility function, we need a computer.

VFI applied to the Buffer Stock Model

Uncertainty brings three new difficulties:

1. **The state space is larger:** loop over all a_{t-1} **and** all z_t
With $\#_a$ and $\#_z$ grid points, that is $\#_a \times \#_z$ maximizations
→ the cost grows exponentially with the number of states: the **curse of dimensionality**
2. z_t **is continuous:** we need to discretize the AR(1)
A routine gives a grid $\mathcal{G}_z$ and a transition matrix Π_z ,
 $\pi_{i_z, i_{z+}} = \Pr[z^{i_{z+}} | z^{i_z}]$
3. **The continuation value is an expectation** $\mathbb{E}_t [v(z_{t+1}, a_t)]$
Costs $\#_z$ interpolations for every trial value of c_t in the maximization

Discretizing an AR(1)

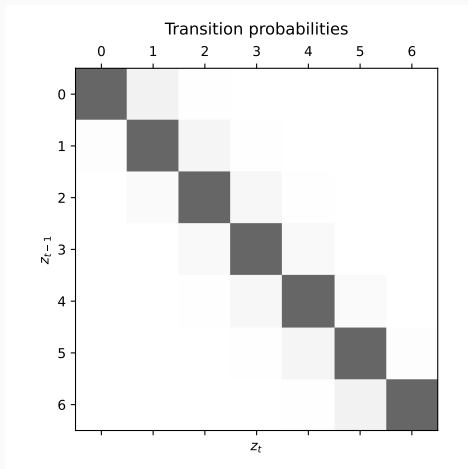
- **Income process:**

$$\log z_t = \rho_z \log z_{t-1} + \psi_t, \quad \psi_t \sim \mathcal{N}(0, \sigma_\psi^2), \quad \mathbb{E}[z_t] = 1$$

- **Tauchen (1986):** evenly spaced grid spanning the unconditional distribution, match the transition probabilities by integrating the normal density
- **Rouwenhorst (1995):** build the transition matrix by recursion, matching the persistence and the unconditional variance *exactly*
 - Preferred at high ρ_z , where Tauchen needs many nodes
- In both cases, it gives us:
 1. A discretized grid for z_t
 2. A Markov transition matrix Π_z with entries $\pi_{i_z, i_{z+}}$

Use built-in functions in **ConSav**: `markov.log_rouwenhorst`,
`markov.tauchen`

Discretizing an AR(1)



Recursive formulation with $\underline{v}$

- One useful trick: precompute the **beginning-of-period value function** (of period $t + 1$, before z_{t+1} is drawn)

$$\underline{v}(z_t, a_t) = \mathbb{E}_t [v(z_{t+1}, a_t)] = \sum_{i_{z+}=0}^{\#_z-1} \pi_{i_z, i_{z+}} v(z^{i_{z+}}, a_t)$$

- Value function:**

$$\begin{aligned} v(z_t, a_{t-1}) &= \max_{c_t} u(c_t) + \beta \underline{v}(z_t, a_t) \\ \text{s.t. } a_t &= (1 + r)a_{t-1} + wz_t - c_t \geq \underline{a} \end{aligned}$$

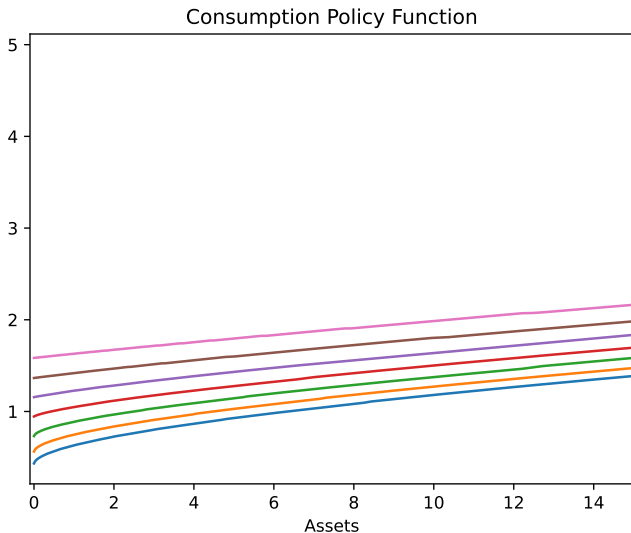
- Why do this?** *Only one interpolation per guess of c_t*

Integrating *inside* the maximization would cost $\#_z$ interpolations per guess.

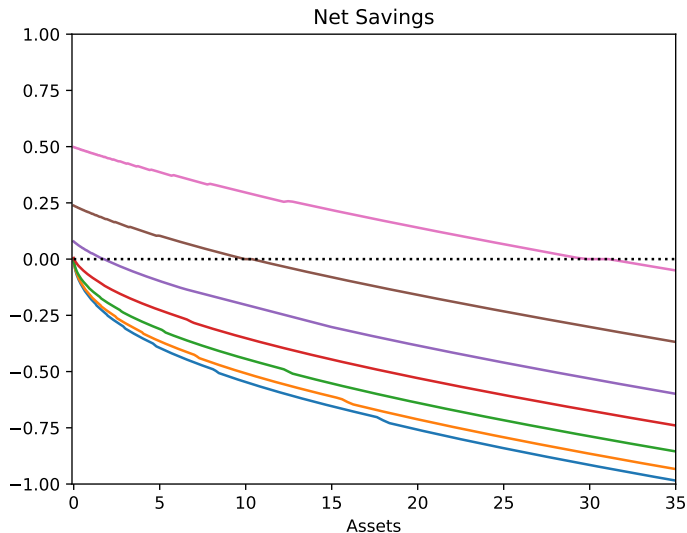
- This is the convention** GEModelTools **uses from the next lecture on**

Let's code!

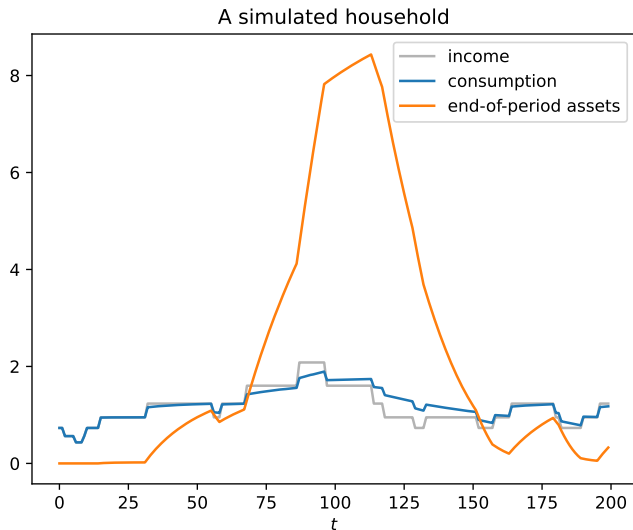
Consumption function

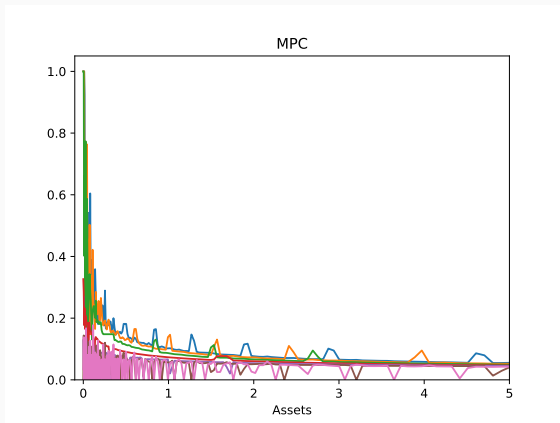


Net savings and the buffer-stock target



One household, simulated





One at the constraint, falling steeply with assets. Under the PIH it was $\frac{r}{1+r}$ for everybody.

- **Consumption lower than under PIH and concave in assets**

Intuition: *Precautionary saving motive is relatively larger for asset poor households because income risk is the same for everybody*

Implications:

1. Windfall gives safety and increases average consumption
⇒ MPC decreasing in assets
2. Attraction towards a buffer-stock target $a_t = a_{t-1}$ despite $\beta R < 1$
3. Larger effective discounting of future income (extreme: no effect of future income changes if constrained before)

EGM



We can do better

- We can do better! We know that an optimal consumption choice must satisfy the Euler equation, so let's use this
- Great thing about this method: don't need a maximization step, just interpolation
- We use information about FOC of the problem: we get a more accurate solution

Overview of EGM

- Assume you have a guess on the consumption function, $c^n(z_t, a_{t-1})$. The envelope condition gives $v_a = (1+r)c^{-\sigma}$, so on a grid of *end-of-period* assets a_t we can compute the marginal beginning-of-period value

$$\underline{v}_a(z_t, a_t) \equiv \mathbb{E}_t [v_a(z_{t+1}, a_t)] = (1+r) \sum_{i_{z+}=0}^{\#_z-1} \pi_{i_z, i_{z+}} c^n(z^{i_{z+}}, a_t)^{-\sigma}$$

- Then back out the state that makes a_t optimal, from the FOC and the budget constraint:

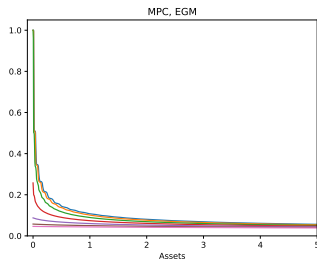
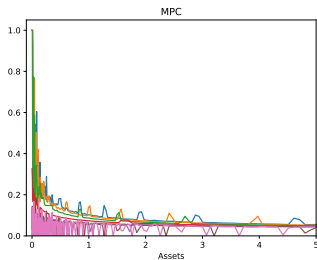
$$\begin{aligned} c_t^{-\sigma} &= \beta \underline{v}_a(z_t, a_t) \\ ((1+r)a_{t-1} + wz_t - a_t)^{-\sigma} &= \beta \underline{v}_a(z_t, a_t) \\ a_{t-1}(z_t, a_t) &= \frac{(\beta \underline{v}_a(z_t, a_t))^{-\frac{1}{\sigma}} + a_t - wz_t}{1+r} \end{aligned}$$

Overview of EGM (continued)

- This is why it's called the **endogenous grid method**: we get an endogenous grid of states a_{t-1} for a fixed grid of choices a_t
- Interpolate to invert the mapping and get $a^*(z_t, a_{t-1})$ on the exogenous grid. If $a^*(z_t, a_{t-1}) < \underline{a}$, set $a^* = \underline{a}$ to enforce the borrowing constraint
- Iterate until convergence, done!

Let's code!

VFI vs EGM



Exercise: permanent productivity types

- **Add permanent types:** $\xi \in \{0.5, 1, 1.5\}$, fixed for life, income $w\xi z_t$
- **Write** `solve_hh_backwards_fix` with the extra leading dimension, `a[i_fix,i_z,:]`, exactly the layout of `household_problem.py` in lecture 04
- **Plot** the consumption functions of the three types
- **Check homogeneity:** with CRRA and $\underline{a} = 0$,

$$c^*(z, a_{-1}; \xi) = \xi c^*(z, a_{-1}/\xi; 1)$$

What does it imply for wealth-to-income ratios across types?

Notebook: 04_egm.ipynb, last section

Simulation and aggregation

From policy functions to aggregates

- **What we have:** $c^*(z_t, a_{t-1})$ and $a^*(z_t, a_{t-1})$, what *one* household does in every state
- **What the next lecture needs:** aggregates, e.g.

$$A_t^{hh} = \sum_{i_z} \sum_{i_{a-}} \mathbf{D}_t(z^{i_z}, a^{i_{a-}}) a^*(z^{i_z}, a^{i_{a-}})$$

where $\mathbf{D}_t$ is the *distribution* of households over $\mathcal{G}_z \times \mathcal{G}_a$

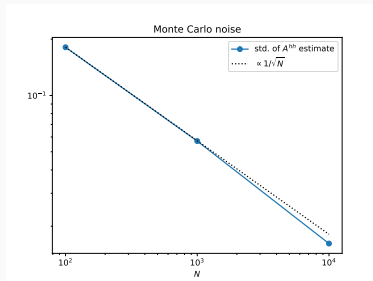
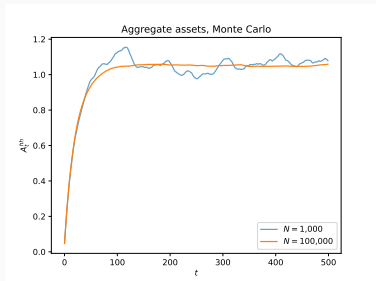
- **So we need to simulate the distribution forward.** Two ways:
 1. Monte Carlo: simulate N households, average
 2. Histogram (Young, 2010): iterate on $\mathbf{D}_t$ itself
- **Notebook:** 05_simulation.ipynb

- **Initial distribution:** draw $z_{i,-1}$ and $a_{i,-1}$ for $i \in \{0, 1, \dots, N-1\}$
- **Simulation:** forwards in time from $t = 0$, in each period
 1. Draw z_{it} given z_{it-1} from the transition probabilities
 2. Use linear interpolation to evaluate

$$a_{it} = a^*(z_{it}, a_{it-1})$$

- **Aggregation:** sample averages, $A_t^{hh} = \frac{1}{N} \sum_i a_{it}$
 - Law of large numbers: correct as $N \rightarrow \infty$
- **Pro:** simple to implement, works for any model
- **Con:** computationally costly, and *random*

Monte Carlo noise



- **Left:** A_t^{hh} never settles, it keeps fluctuating around its stationary level
- **Right:** the error falls like $1/\sqrt{N}$: quadrupling N halves it
- **Why this hurts:** next lecture we root-find on asset market clearing, $A^{hh}(r) - K(r) = 0$. A stochastic A^{hh} puts a floor on the achievable tolerance

The distribution as an object

- **Idea:** stop tracking households, track *probability mass*
- **Histogram:** an array $\mathbf{D}$ on $\mathcal{G}_z \times \mathcal{G}_a$, summing to 1
 - $\underline{\mathbf{D}}_t$: beginning-of-period, over (z_{t-1}, a_{t-1})
 - $\mathbf{D}_t$: after the productivity draw, over (z_t, a_{t-1})
- **The productivity transition is exact:**

$$\mathbf{D}_t(z^{i_z}, a^{i_a-}) = \sum_{i_{z-}} \pi_{i_{z-}, i_z} \underline{\mathbf{D}}_t(z^{i_{z-}}, a^{i_a-}) \iff \mathbf{D}_t = \Pi'_z \underline{\mathbf{D}}_t$$

- **The problem:** all mass at (z^{i_z}, a^{i_a-}) saves $a^*(z^{i_z}, a^{i_a-})$, which falls *between* grid points

The savings lottery

- **Split the mass between the two neighbors.** Find i such that $a^i \leq a^* \leq a^{i+1}$ and set

$$\omega = \frac{a^{i+1} - a^*}{a^{i+1} - a^i} \in [0, 1]$$

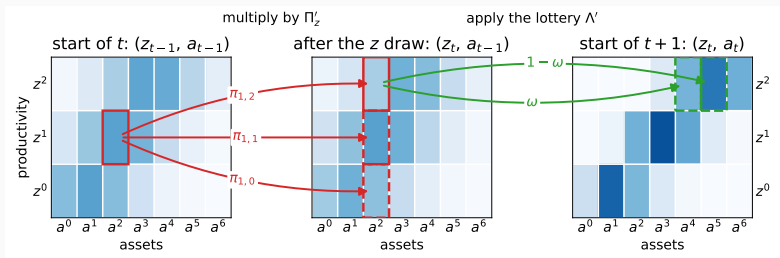
- **Assign** share ω to a^i and $1 - \omega$ to a^{i+1}
 - The lottery has mean *exactly* a^* : aggregate savings are not distorted by the grid
- **One full period:**

$$\underline{D}_t \xrightarrow{\Pi'_z} D_t \xrightarrow{\Lambda'} \underline{D}_{t+1}$$

where Λ collects the lottery weights

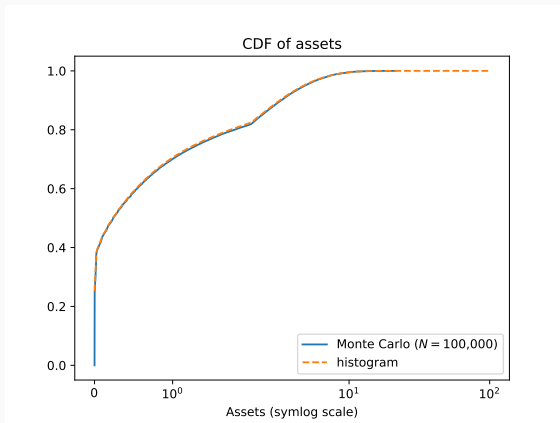
- **Both maps are linear and deterministic:** simulation becomes matrix algebra. No draws, no noise

One period forward, in pictures



- **Productivity draw:** the mass at (z^1, a^2) is spread over the column a^2 , with the weights of row 1 of Π_z . Assets do not move
- **Savings lottery:** those who drew z^2 save $a^* \in [a^4, a^5]$. Their mass moves along the row, split ω and $1 - \omega$. Productivity does not move
- **Every cell does both**, and the two maps are the matrices Π'_z and Λ'

Monte Carlo vs histogram



Same distribution. The histogram is the exact limit the Monte Carlo approaches by brute force, at a fraction of the cost. Its only cost: the distribution lives *on the grid*.

- **Aggregates are sums over the grid:**

$$A^{hh} = \sum_{i_z, i_a} D(z^{i_z}, a^{i_a}) a^*(z^{i_z}, a^{i_a}), \quad C^{hh} = \sum_{i_z, i_a} D c^*$$

- **Stationary distribution:** iterate the forward step until D stops moving (526 iterations, 0.01 secs)

Exercise: death

- **Perpetual youth:** after saving, die with probability $\delta = 0.02$.
Replaced by a newborn with $a = 0$ and ergodic z . No annuities, the assets of the dead are lost.
- **Backward:** $\beta \rightarrow \beta(1 - \delta)$. No new code, use `par._replace`
- **Forward:** add a death step to `simulate_hh_forwards`. Where exactly?
- **Compare** the CDF of assets and A^{hh} with the baseline
- $C^{hh} = rA^{hh} + w$ **now fails**. Derive the identity that replaces it and check it

Notebook: 05_simulation.ipynb, last section

Recap

Recap I: buffer-stock theory

- **PIH:** consume the annuity value of total wealth, $MPC = \frac{r}{1+r} \approx 2\%$, consumption linear in wealth, the distribution is irrelevant. **Data:** 40-50%.
- **Two ingredients fix it:** uninsurable income risk and a borrowing constraint.
- **Euler-equation with an occasionally binding constraint:**

$$u'(c_t) \geq \beta R \mathbb{E}_t[u'(c_{t+1})], \quad \text{with equality if } a_t > \underline{a}$$

- **Prudence** ($u''' > 0$) $\times$ **risk** $\Rightarrow$ **precautionary saving:** consume less today, tilt consumption into the future.
- **Policy functions:** consumption *concave* in assets. MPC one at the constraint, falling with wealth. Net savings cross zero: a **buffer-stock target**, even with $\beta R < 1$.
- **Consequence:** who holds what matters. The aggregate MPC is a property of the *distribution*.

Recap II: EGM

- **VFI:** an optimizer at every grid point, interpolating the *value* function. Slow and ragged policies.
- **EGM:** iterate on the Euler-equation, with the grid on the *choice* a_t instead of the state a_{t-1} . Given $\underline{v}_a(z_t, a_t)$:
 1. Invert the FOC: $c(z_t, a_t) = [\beta \underline{v}_a(z_t, a_t)]^{-1/\sigma}$
 2. Endogenous grid from the budget: $a_{t-1} = \frac{a_t + c - wz_t}{1+r}$
 3. Interpolate a^* back on the fixed grid, impose $a^* \geq \underline{a}$, get c^* from the budget
 4. Envelope + expectation: $\underline{v}_a = \Pi_z [(1+r)(c^*)^{-\sigma}]$
- **No optimizer, one interpolation per z :** $\approx 15\times$ faster per step, same number of steps (contraction with modulus β).
- **Smoother:** interpolates the *policy* function, which is close to linear. Clean MPCs.
- `solve_hh_backwards` is the backward step of `GEModelTools`.

Recap III: the histogram method

- **To aggregate we need the distribution D** over $\mathcal{G}_z \times \mathcal{G}_a$.
- **Monte Carlo:** simulate N households. Burn-in, and noise $\propto 1/\sqrt{N}$ that never goes away: fatal for root-finding on market clearing.
- **Histogram (Young, 2010):** track probability mass, not households.

Two deterministic transitions:

1. Productivity: $D_t = \Pi'_z D_t$
 2. Savings lottery: $a^* \in [a^i, a^{i+1}]$, share $\omega = \frac{a^{i+1} - a^*}{a^{i+1} - a^i}$ to a^i , $1 - \omega$ to a^{i+1} . Mean-preserving, so aggregate savings are not distorted.
- **Iterate until D stops moving:** the stationary distribution. Exact given the grid, no random numbers, a fraction of the time.
 - `simulate_hh_forwards` is the forward step of `GEModelTools`.

Recap IV: aggregation

- **Aggregates are sums over the grid:**

$$A^{hh} = \sum_{i_z} \sum_{i_{a-}} \mathbf{D}(z^{i_z}, a^{i_{a-}}) a^*(z^{i_z}, a^{i_{a-}}), \quad C^{hh} = \sum \mathbf{D} c^*$$

with $\mathbf{D}$ *after* the productivity draw, so it lines up with the policy functions.

- **Check:** in the stationary distribution $C^{hh} = rA^{hh} + w$.
- **Aggregate MPC:** $\sum \mathbf{D} \text{MPC} \approx 0.33$, against 0.02 under the PIH. A quarter of households sit at the constraint.
- **Same recipe next lecture**, but r and w will be chosen so that A^{hh} clears the asset market.

The household block in practice

1. **Backward step (EGM):** iterate

$$\underline{v}_a \rightarrow c^*(z_t, a_{t-1}), a^*(z_t, a_{t-1})$$

until the policy functions converge $\Rightarrow$ solve_hh_backwards

2. **Forward step (histogram):** build the lottery from a^* , iterate

$$\underline{D}_t \xrightarrow{\Pi'_z} D_t \xrightarrow{\Lambda'} \underline{D}_{t+1}$$

until the distribution converges $\Rightarrow$ simulate_hh_forwards

3. **Aggregate:** $A^{hh} = \sum D a^*$, $C^{hh} = \sum D c^*$

This pair is the household block. Next lecture GEModelTools runs it for us, and r and w become equilibrium objects.

What you should remember

1. **The PIH gives an MPC of about 2%; the data say 40 – 50%**
2. **Uncertainty plus a borrowing constraint fixes it**
3. **EGM solves the household problem fast and accurately**
4. **The histogram method gives a precise distribution, without simulation noise**

Next: *Stationary equilibrium*

You should:

1. Study the code from this lecture
2. Glance at Aiyagari (1994), “Uninsured Idiosyncratic Risk and Aggregate Saving”

Exercise: endogenous labor supply

$$v(z_t, a_{t-1}) = \max_{c_t, n_t} \frac{c_t^{1-\sigma}}{1-\sigma} - \varphi \frac{n_t^{1+\nu}}{1+\nu} + \beta \mathbb{E}_t [v(z_{t+1}, a_t)]$$
$$\text{s.t. } a_t = (1+r)a_{t-1} + wz_t n_t - c_t \geq 0$$

1. **FOCs:** show that $\varphi n_t^\nu = wz_t c_t^{-\sigma}$, constrained or not
2. **EGM, unconstrained:** Euler gives c , the intratemporal FOC gives n , the budget gives a_{t-1} . One extra line
3. **EGM, constrained:** $a^* = 0$, but c and n must be found together.
Newton on one equation in n
4. **Policy functions:** does labor supply fall with wealth? With z ?
5. **Simulate and aggregate:** $L^{hh} = \sum D z n^*$, check $C^{hh} = rA^{hh} + wL^{hh}$. Share at the constraint?

Notebook: 06_egm_endogenous_labor.ipynb

Misc



1. Life-cycle (I)

- **Basically:**
 1. Born, working, retired, die
 2. Age-varying parameters (esp. income)
- **Add-ons:**
 1. Labor supply, human capital, occupation
 2. Portfolio choice and entrepreneurship
 3. Family formation
 4. Health, mortality
 5. etc.
- **Good starting example:** “Life-Cycle Consumption and Children: Evidence from a Structural Estimation”, Jørgensen (2017)

1. Life-cycle (II)

Paper: Gourinchas and Parker (2002)

Life-cycle consumption-saving model with retirement

- Young households:
Save for precautionary reasons (buffer)
- Older households:
Save for retirement (life-cycle)

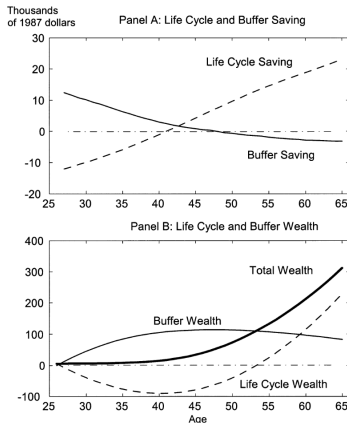
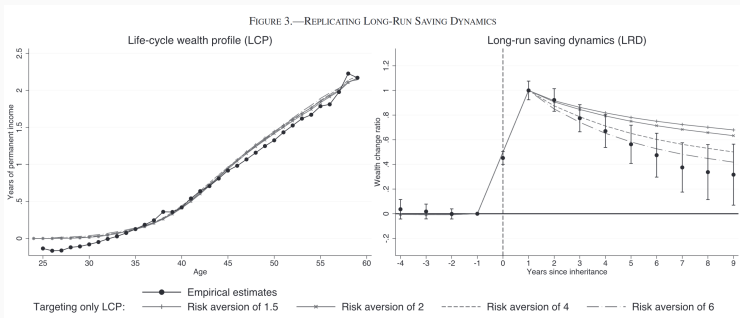


FIGURE 7.—The role of risk in saving and wealth accumulation.

1. Life-cycle (III)

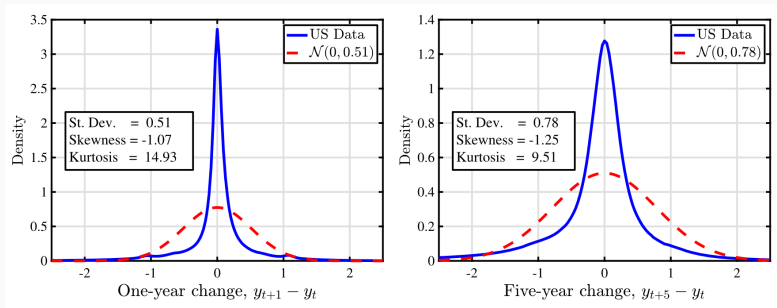
- **Natural experiment:** Wealth depletion after sudden inheritance
- **Results:**
 1. Life-cycle profile of wealth fitted for many levels of risk-aversion (by varying the discount factor)
 2. Fast wealth depletion requires high risk-aversion (or high perceived risk)



Source: Druedahl and Martinello (2022)

2. More realistic income risk (I)

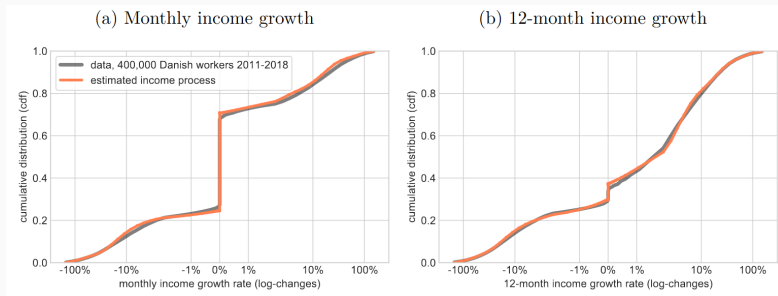
Annual earnings-changes are far from log-normal:



Source: Guvenen et al. (2021)

2. More realistic income risk (II)

Many with zero-growth month-month:



Source: Druedahl et al. (2021)

3. Epstein-Zin

$$\begin{aligned} v_t(z_t, m_t) = & \max_{c_t} \left[(1 - \beta) \cdot c_t^{1-\sigma} + \beta \cdot w_{t+1}^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \\ \text{s.t.} \quad & w_{t+1} \equiv \mathbb{E}_t \left[v_{t+1}(z_{t+1}, m_{t+1})^{1-\gamma} \right]^{\frac{1}{1-\gamma}} \\ & m_{t+1} = (1 + r)(m_t - c_t) + y_{t+1} \end{aligned}$$

- **Preferences:**

1. Patience: β
2. Intertemporal substitution: σ
3. Risk-aversion: γ

- **Euler-equation:** $c_t^{-\sigma} = \beta R \cdot \mathbb{E}_t \left[c_{t+1}^{-\sigma} \cdot \left(\frac{w_{t+1}}{v_{t+1}} \right)^{\gamma-\sigma} \right]$

1. FOC: $0 = v_t^\sigma \cdot \left[(1 - \beta) \cdot c_t^{-\sigma} - \beta R \cdot w_{t+1}^{\gamma-\sigma} \cdot \mathbb{E}_t \left[v_{t+1}^{-\gamma} \cdot \frac{\partial v_{t+1}}{\partial m_{t+1}} \right] \right]$
2. Envelope condition: $\frac{\partial v_t(z_t, m_t)}{\partial m_t} = v_t^\sigma \cdot (1 - \beta) \cdot c_t^{-\sigma}$

4. Deep learning

- **Curse of dimensionality:**
 1. Many states
 2. Many choices
 3. Many shocks
- **Deep (reinforcement) learning:**
 1. Approximate value and policy functions with *neural networks*
 2. Approximate on simulation sample rather than on grid
 3. Automatic differentiation (backpropagation) and GPUs for speed
- **Examples:** Maliar and Maliar (2021) and Azinovic and Scheidegger (2022)
- **Working paper:** Druedahl and Røpke (2025)
Python package: EconDLSolvers